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Spatiotemporal intermittency regimes of the one-dimensional complex Ginzburg–Landau equation

Hugues Chaté

Service de Physique de l'Etat Condensé, Centre d'Etudes de Saclay, 91191 Gif-sur-Yvette Cedex, France

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Abstract. Completing an earlier work, the 'phase diagram' of the one-dimensional complex Ginzburg–Landau equation is presented. In the Benjamin–Feir stable region, spatiotemporal intermittency regimes are identified which consist of patches of linearly stable plane waves separated by localized objects with a well defined dynamics. The simplest of these structures are shown to be members of a family of exact solutions discovered by Nozaki and Bekki. The problem of the determination of the parameter domain of existence of spatiotemporal intermittency is discussed. In particular, given the inadequacy of the quantities usually measured to determine spatiotemporal disorder, only a rather crude determination of the limit of spatiotemporal intermittency is proposed, awaiting further knowledge on the nature, stability and interactions of the localized objects involved. In the transition region, asymptotic states with an irregular, frozen spatial structure are shown to occur. Finally, the disordered regimes observed in the Benjamin–Feir unstable region are reviewed and argued to be of the spatiotemporal intermittency type in some cases.

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1. Introduction

In a recent paper [1], the spatiotemporally chaotic regimes exhibited by the one-dimensionally complex Ginzburg–Landau equation in the so-called Benjamin–Feir unstable region of its parameter space were presented. Here, I give a detailed account of the spatiotemporally disordered regimes observable in the complementary part of the parameter space (Benjamin–Feir stable region) where there exists a band of wavevectors corresponding to stable plane wave solutions. These results allow a comprehensive discussion of the disordered regimes in the complete parameter space.

The general setting remains the same: in the infinite-size, 'thermodynamic' limit, weak turbulence models such as the complex Ginzburg–Landau equation show *extensive chaos*, as proved by numerical studies and mathematical considerations [2]. This calls, in turn, for an understanding of spatially extended nonequilibrium systems in terms of concepts akin to those developed in statistical mechanics and in the field of critical phenomena. As a matter of fact (as will be argued below), the disordered regimes considered in the present study are typical of *spatiotemporal intermittency*, a phenomenon occurring in many different numerical and experimental situations for which the validity of a description inspired by critical phenomena has been shown to be particularly good [3].

The complex Ginzburg–Landau equation describes the modulations of a spatially extended system which has undergone a bifurcation to an oscillatory state; it is thus very

general, as shown from the analysis of many physical, chemical, and biological situations. Here only its simplest version in one space dimension is considered, thereby hopefully focusing on the generic properties of the model. Suitable scaling of the complex amplitude A , space, and time, allows the equation to be written in function of only two real parameters b_1 and b_3 :

$$\partial_t A = A + (1 + ib_1)\partial_{xx} A - (b_3 - i)|A|^2 A$$

where $b_3 > 0$ and $x \in [0, L]$. This somewhat unusual choice of scaling is related to the region of the parameter space that is of interest here [4]. The 'real' Ginzburg-Landau equation, known to possess a purely relaxational dynamics [5], is recovered in the $b_1 = 0, b_3 \rightarrow \infty$ limit. Another limit of interest is $b_3 = 0, b_1 \rightarrow \infty$, corresponding to the integrable, purely dispersive dynamics of the nonlinear Schrödinger equation [6]. Plane wave solutions of the form $A = a_k \exp i(kx + \omega_k t)$ with $a_k^2 = (1 - k^2)/b_3$ and $\omega_k = 1/b_3 - (b_1 + 1/b_3)k^2$ are linearly stable, if $b_1 < b_3$ and $k^2 < k_{\text{Eckhaus}}^2 = (b_3 - b_1)/(3b_3 - b_1 + 2/b_3)$ [7]. This band of wavevectors shrinks to zero on the $b_1 = b_3$ (BF) line, which corresponds to the Benjamin-Feir or modulational instability of the $k = 0$ state.

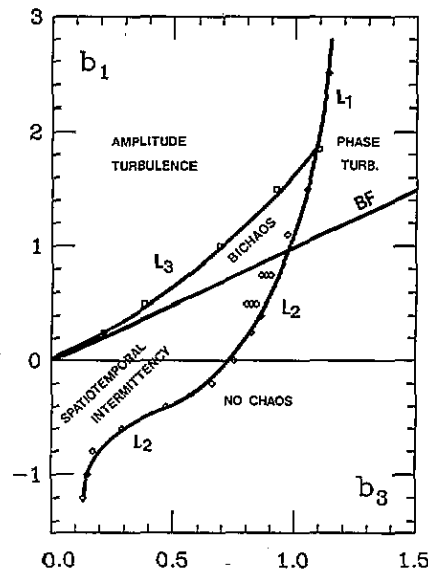


Figure 1. Phase diagram in the (b_1, b_3) parameter plane. Line L_1 and the portion of line L_2 above the BF Benjamin-Feir line were determined in [1]. Details on the determination of line L_3 and in particular when $b_3 \rightarrow 0$ will be reported elsewhere [26]. The points of line L_2 for $b_1 = 0.5$ and $b_1 = 0.75$ give an idea of the importance of the size-effects when approaching the BF line. The rightmost points were obtained for the largest sizes ($L = 1000, 2000$).

It was shown in [1] that the complex Ginzburg-Landau equation exhibits spatiotemporally disordered regimes ('amplitude' or 'defect' turbulence) in the region of the (b_1, b_3) plane to the left of lines L_1 and L_2 (figure 1). In particular the line L_1 was determined via the measure of quantities such as correlation functions and the density of defects. In the present paper, the prolongation of L_2 below the Benjamin-Feir $b_1 = b_3$ line is studied. For

$b_1 > b_3$, the equation also exhibits ‘phase turbulence’ regimes well-described by a phase equation of the Kuramoto–Sivashinsky type [8]. This dynamics, during which $|A|$ never reaches zero, is observed up to the lines L_1 and L_3 (figure 1), leaving thus a ‘bi-chaotic’ region delimited by the BF, L_1 and L_3 lines in which, depending on the initial conditions, one may observe either amplitude of phase turbulence.

The linear stability of plane wave solutions with $k^2 < k_{\text{Eckhaus}}^2$ for $b_1 < b_3$ does not exclude the existence of other non-trivial solutions coexisting at the same parameter values. Understandably, phase turbulence is not observed in the Benjamin–Feir stable region but, *a priori*, nothing rules out amplitude turbulence regimes or even non-chaotic nonlinear solutions. Indeed, as shown below, the linearly-stable plane waves are unstable with respect to finite-amplitude perturbations introduced, say, in the initial conditions, giving way to sustained regimes of spatiotemporal intermittency in a well-defined region of the parameter space.

2. Disordered regimes in the Benjamin–Feir stable region

Most of the numerical simulations reported here use a pseudo-spectral code with a ‘slaved-frog’ [9] temporal scheme and periodic boundary conditions. Except for special runs aimed at investigating details of the dynamics, the spatial resolution is $\delta x \simeq 1$ and the time-step δt varies from 0.01 to 0.2, depending on the value of b_3 (ω_k provides the relevant timescale). Systems of size $L = 500$ to 4000 have been considered and found large enough to approximate the ‘thermodynamic limit’ in most cases. To avoid the stable plane wave solutions present for $b_1 < b_3$, some disorder in the initial conditions is needed. In practice, initial conditions consisting of a rather steep pulse of amplitude proved sufficient to generate the solutions presented below.

For b_3 small enough, sustained spatiotemporally disordered regimes are easily generated. The system settles in a statistically well-defined regime after a rather short, size-independent, transient. For larger values of b_3 , any disorder decays to a pure plane wave solution (most often a standing wave ($k = 0$)). The problem then arises of defining and determining the transition line L_2 delimiting the domain of existence of spatiotemporal disorder. With this aim, I first describe and analyse the spatiotemporal intermittency regimes observed in the region of interest.

Figures 2–5 show four typical disordered regimes observed in an $L = 500$ system with periodic boundary conditions after transients have been discarded. These four examples provide a fairly comprehensive picture of the disordered regimes in the Benjamin–Feir stable region. They share the same main characteristics: localized structures evolve in a complex spatiotemporal manner, separating larger regions of almost-constant amplitude which are nothing but patches of (stable) plane wave solutions. All the wavenumbers measured in these large regions have been found to be within the linearly stable band. In figure 2, only propagative and branching objects are observed, leaving triangular embeddings of standing waves ($k \simeq 0$) where $|A| \simeq a_0 = 1/\sqrt{b_3}$. Figure 3 shows various hole-like and shock-like objects with quasi-constant velocities separating large regions of plane wave solutions. Figure 4 can be seen as a mixture of the two previous cases, indicating that very complex dynamics may be observed. In figure 5, the spatial extension of the system is broken by local, irregular arrangements of stationary hole-like and shock-like objects between which a more turbulent dynamics takes place.

All these regimes, and especially those of figure 2, are reminiscent of the well-known spatiotemporal patterns generated by class 3 cellular automata for which the triangular

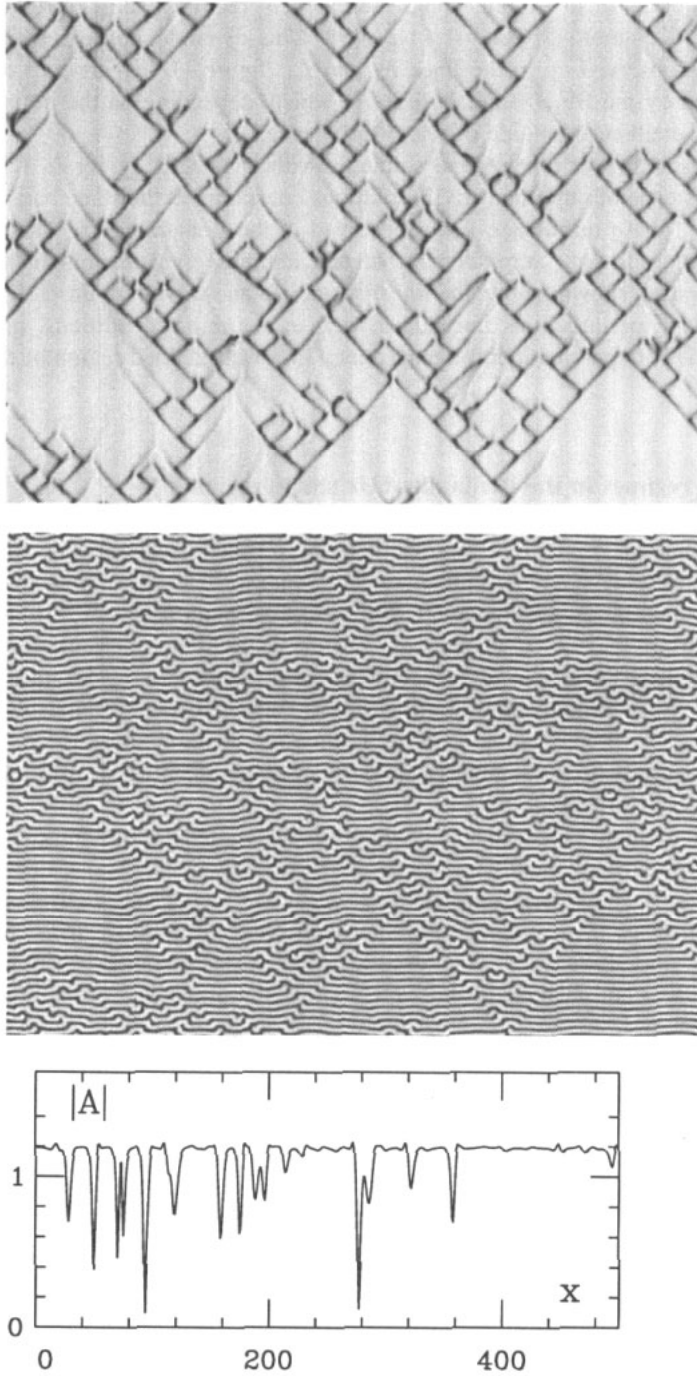


Figure 2. Spatiotemporal intermittency regime observed for $b_1 = 0.5$, $b_3 = 0.7$, during $T \simeq 330$. Spacetime diagrams with time running upward of (a) grey level representation of $|A|$ from $|A| = 1.25$ (white) to $|A| = 0$ (black); (b) grey level representation of the phase from $\phi = 0 \bmod(2\pi)$ (white) to $\phi = \pi \bmod(2\pi)$ (black). The modulus of the initial solution is shown in (c). Note the branching process occurring almost periodically at spacetime points where a π phase-slip is observed.

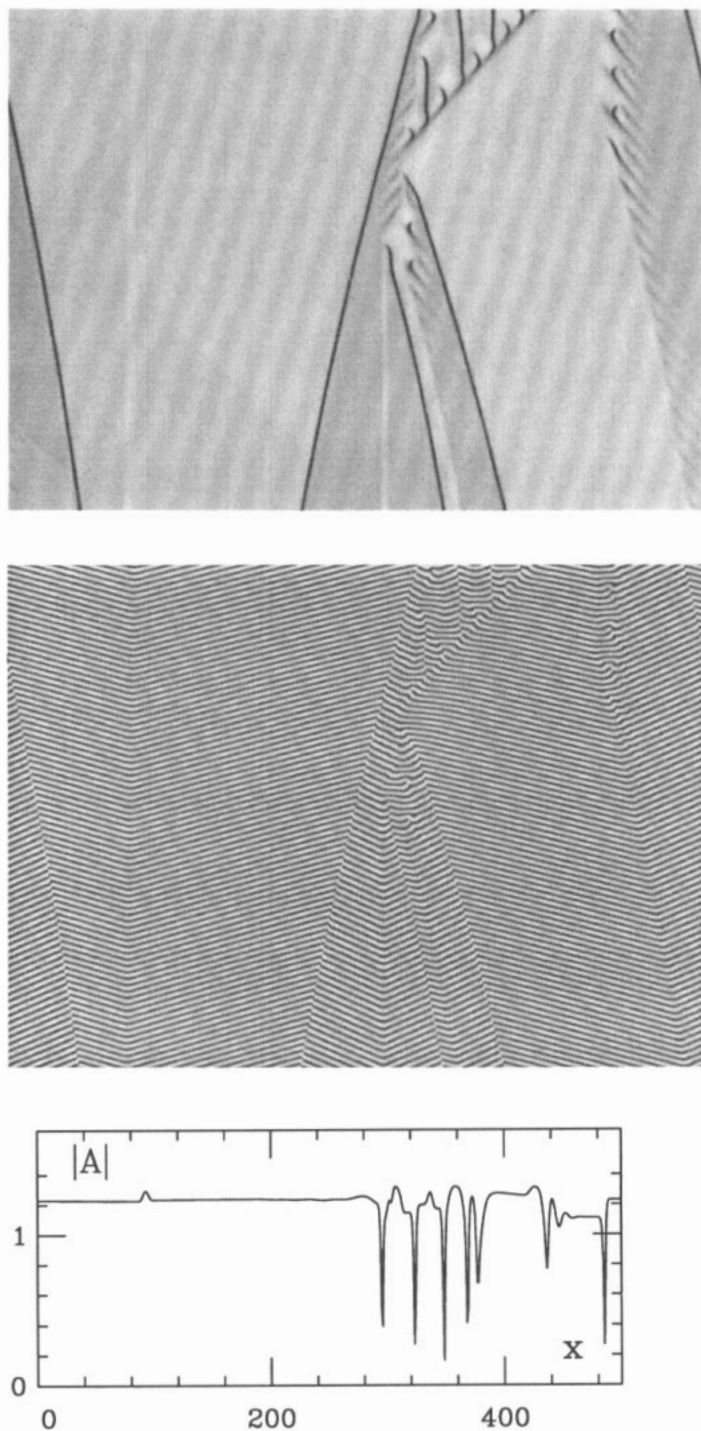


Figure 3. Same as figure 2 but for $b_1 = -0.25$, $b_3 = 0.57$. Part (a): the grey scale goes from $|A| = 1.35$ (white) to $|A| = 0$ (black). Note the diversity of localized structures (holes, shocks, etc) and the regularity of the plane waves separating them (b). Note also the phase jump across each localized object. Part (c) shows the modulus of the final solution of the run.

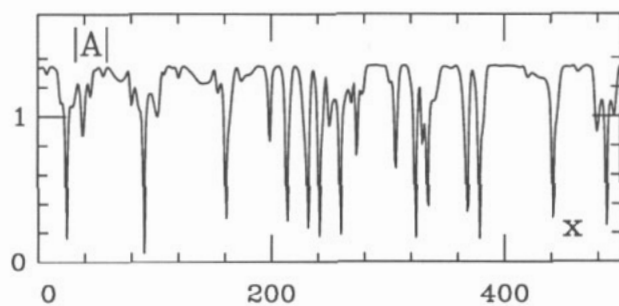
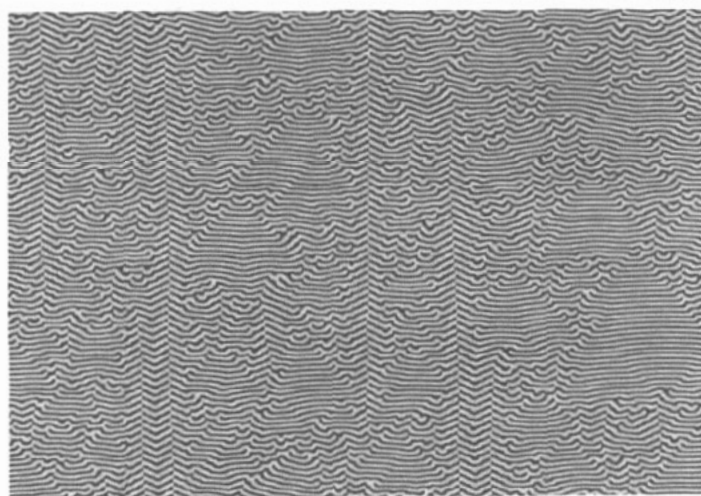
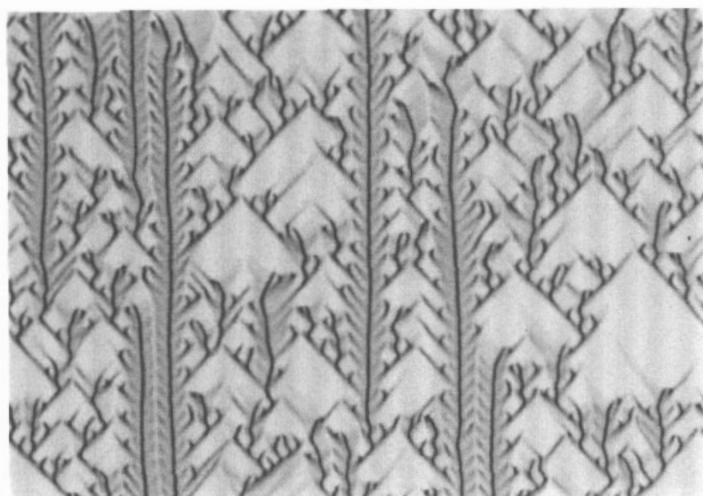


Figure 4. Same as figure 2 but for $b_1 = 0$, $b_3 = 0.55$. The white level in part (a) is $|A| = 1.35$. Note the long-lived hole-like objects with small velocity and a phase jump across them constantly close to π .

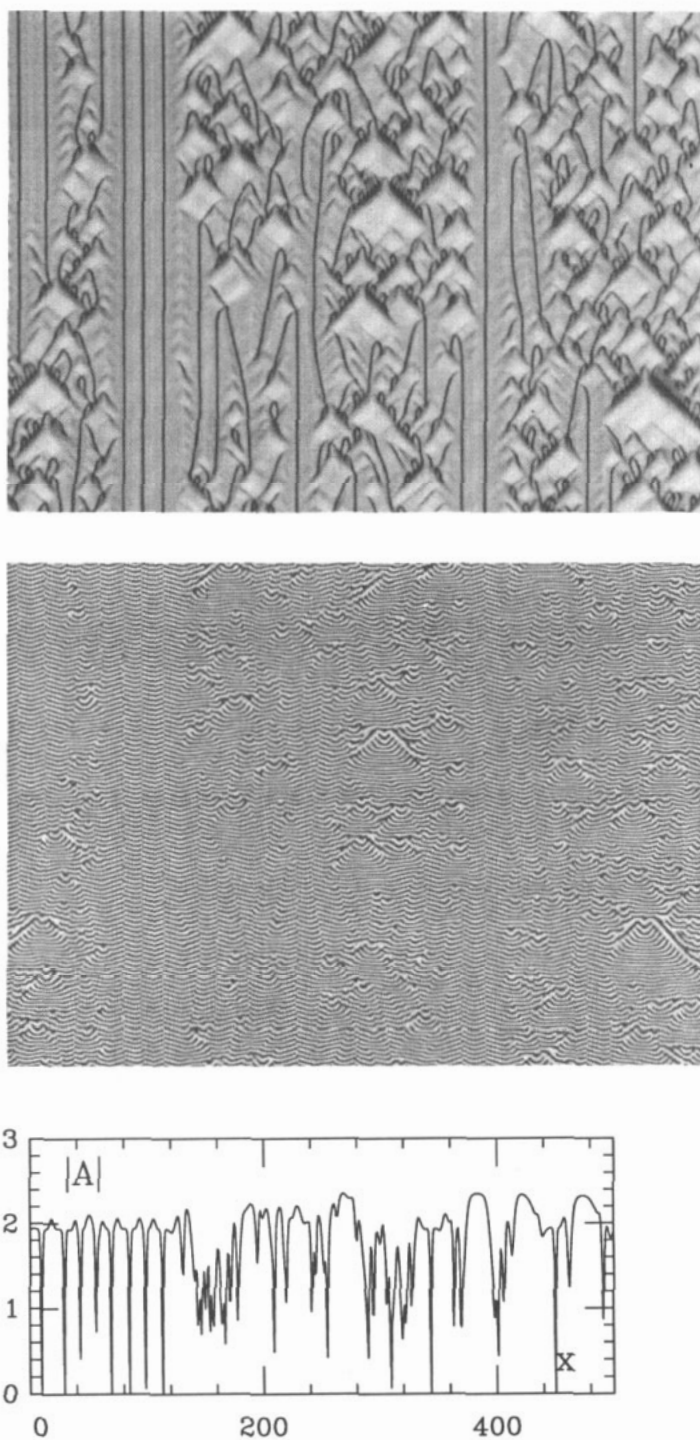


Figure 5. Same as figure 2 but for $b_1 = -0.75$, $b_3 = 0.18$. Time is running upward for $T = 165$. The maximum level in (a) is $|A| = 2.37$. Part (c) shows the initial solution of the run. Note the irregular arrangement of holes and shocks.

embeddings are formed of sites in an absorbing state [10]. The plane waves, which make up the regular regions, play exactly this role: they are linearly stable and thus do not spontaneously nucleate another phase. The propagation of spatiotemporal disorder occurs only *via* a contamination process by the localized objects. The disordered regimes observed for $b_1 < b_3$ are therefore determined to be *spatiotemporal intermittency regimes stricto sensu*, since, in addition to the mere observation of the intermittent character of the solutions in both time and space, one is assured of the absorbing character of the regular, 'laminar' state [3]. To be more precise, one should actually speak of a continuum of absorbing states since every linearly stable plane wave solution acts as such: taken as an initial condition, or for initial conditions in its basin of attraction, it will never spontaneously break down to generate the localized, defect-like objects.

The types of regimes presented in figures 2–5 exist in layer-like regions of the (b_1, b_3) plane, leading to different-looking transitions when varying b_3 at fixed values of b_1 . For example, only regimes of the type presented in figure 2 are found when $b_1 \gtrsim 0.2$. Increasing b_3 , the sizes of the triangular 'laminar' domains increase, and, eventually, one is left with a few isolated objects. For smaller values of b_1 , these 'triangles-only' regimes are progressively replaced by regimes similar to those of figures 3 and 4. For $b_1 \lesssim -0.5$, the transition occurs via the increasing importance of spatially-disordered but stationary arrangements of hole-like and shock-like objects in an otherwise strongly turbulent dynamics (see figure 5). As the average lifetime of the hole-like structures diverge upon increasing b_3 , the whole system may eventually be covered by an irregular arrangement of these objects for some range of b_3 (figure 6).

If, for b_3 large enough, one always recovers a pure plane wave solution, a rich variety of objects is revealed in the transition region where the 'freezing' of the spatiotemporally intermittent regimes occur (see, e.g. figures 7, 8).

3. Localized objects in the Benjamin–Feir stable region

In [1], a quantitative analysis in term of density of defects was developed to determine the transition line L_1 . To assert the adequacy of such an approach for the region $b_1 < b_3$, it is necessary to know to what extent the localized objects observed are 'real' defects (i.e. zeros of A), especially those, frequent, for which $|A|$ remains small in their core (figures 3, 4). Contrary to the two-dimensional case, there is no topological reason, in one space dimension, for which defects should be stable and disappear only by pair annihilation [11]. This apparent contradiction is dissipated in figure 9, where a high-resolution run is presented which shows that the $\text{Re}(A) = 0$ and $\text{Im}(A) = 0$ lines 'condense' along the trajectory of the hole-like objects but cross each other only at scattered points in space-time. Thus, in accord with topological arguments, true defects do not have a finite lifetime. Yet the hole-like objects possess the remarkable property of keeping a very small amplitude associated to a sharp phase jump in their core.

Figure 10 reveals the structure of a hole-like object with a finite velocity. No true defects are observed, as the $\text{Re}(A) = 0$ and $\text{Im}(A) = 0$ lines condense on a space-time line but never cross each other. The variety of velocities observed for the hole-like (and shock-like) objects is reminiscent of the family of exact solutions found by Nozaki and Bekki [12, 13]. Though the complex Ginzburg–Landau equation possesses the Painlevé property only in the $(b_1 \rightarrow \infty, b_3 = 0)$, nonlinear Schrödinger equation limit [14], Nozaki and Bekki [12, 13], and later Cariello and Tabor [15], found exact solutions with methods related to Painlevé analysis. Recently, Conte and Musette have shown that most exact solutions of

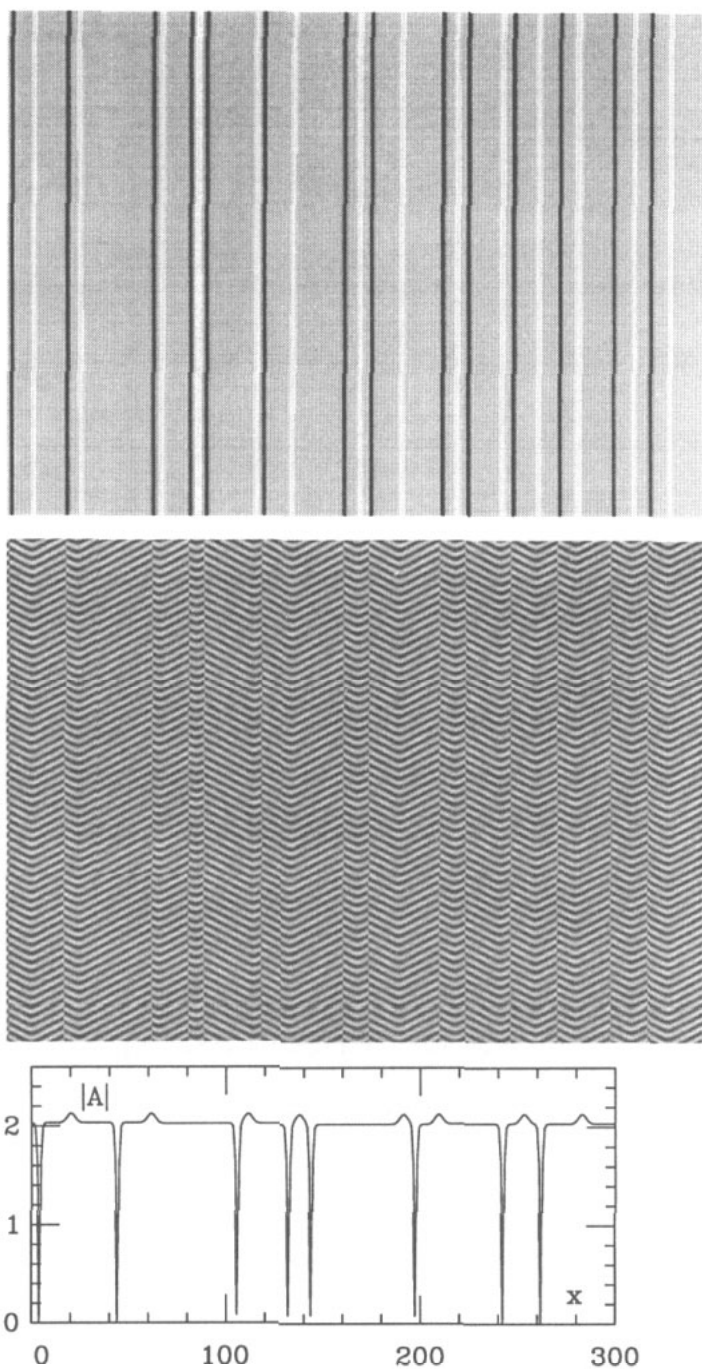


Figure 6. Regular asymptotic state obtained after spatiotemporal disorder has died out. This stationary, irregular arrangement of holes and shocks was obtained for $b_1 = -0.9$, $b_3 = 0.18$ on a system of size $L = 500$. Time is running upward for $T = 82.5$. Same representations as in figures 2–5. Part (a): grey levels of $|A|$ with maximum level (white) at $|A| = 2.15$. Part (b): grey levels of the phase. Part (c): profile of the modulus of a part of the solution. Note that some holes, closely connected to shocks, do not reach $|A| = 0$, forming ‘bound states’. The isolated-enough objects are $c \approx 0$ Nozaki–Bekki solutions.

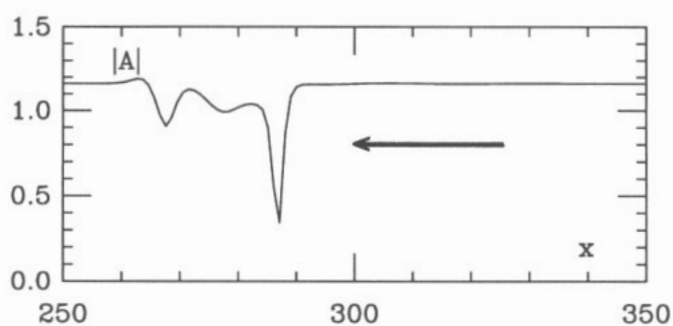
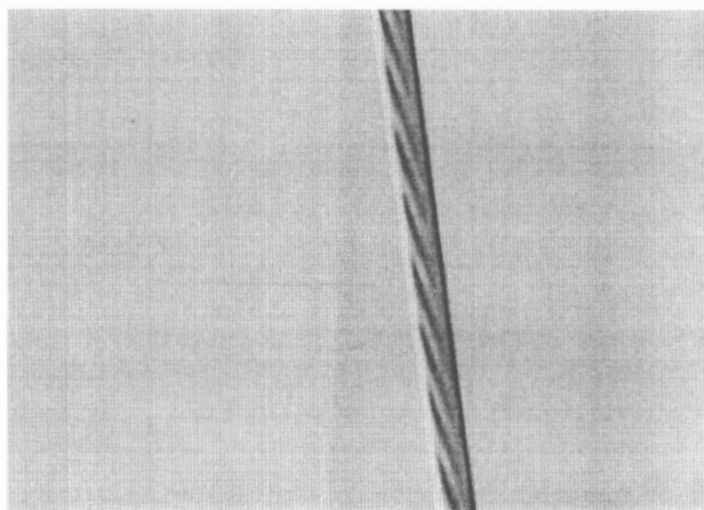


Figure 7. Same as figure 6 but for a propagating, oscillating object obtained for $b_1 = -0.2$, $b_3 = 0.67$ in a $L = 500$ system. Time is running upward for $T = 165$. The speed of the object is $c \simeq 0.30$. The maximum level in (a) is $|A| = 1.25$.

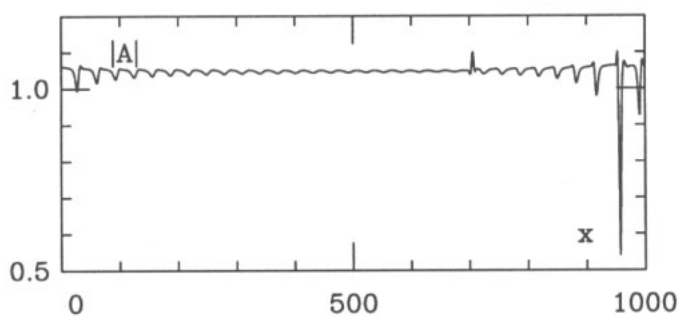
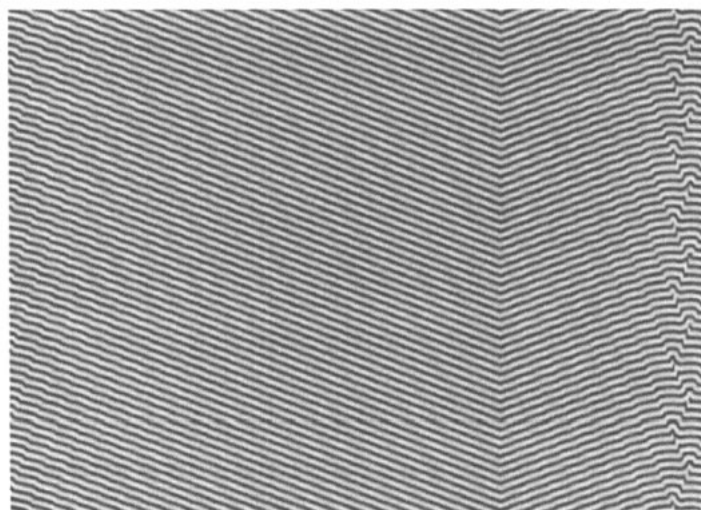
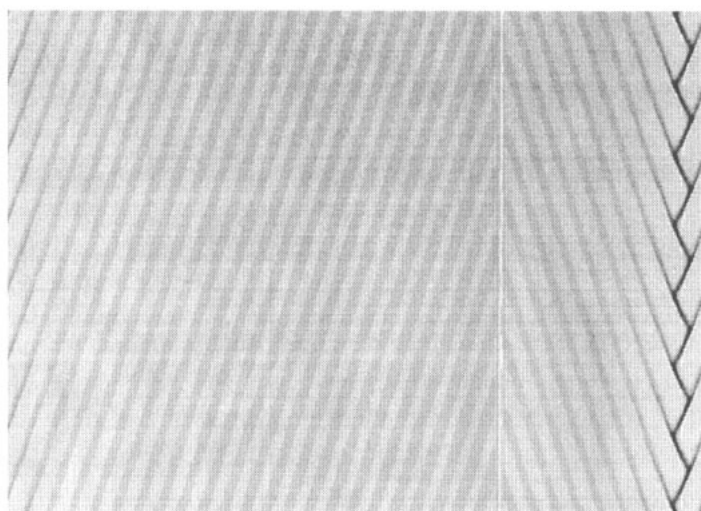


Figure 8. Same as figure 6 but for $b_1 = 0.75$, $b_3 = 0.88$, $L = 1000$. Time is running upward during $T = 330$. The maximum level in (a) is $|A| = 1.12$.

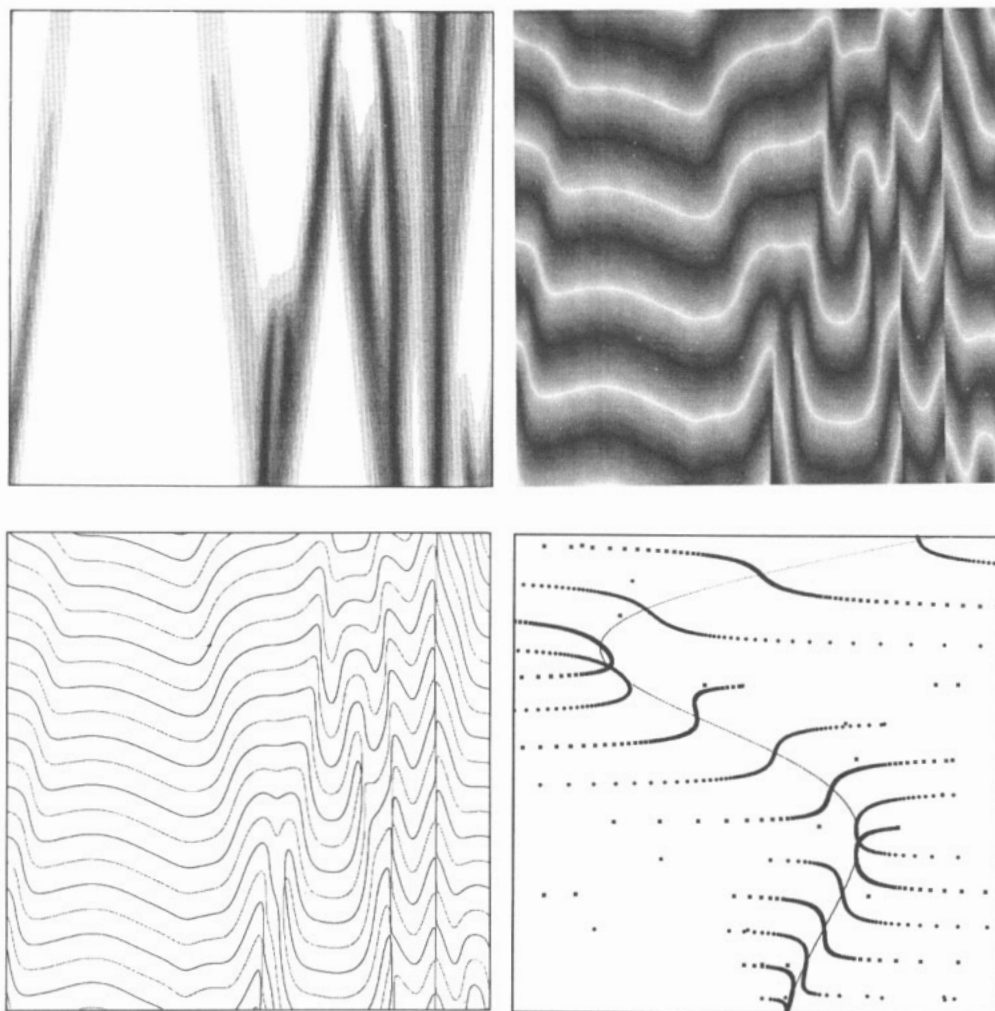


Figure 9. High-resolution run of a system of size $L = 125$ for $b_1 = 0$ and $b_3 = 0.6$. Time is running upward during $T = 18$. Part (a) is a spatiotemporal grey level plot of $|A|$. Part (b) is the corresponding representation of $\arg(A)$. Part (c) shows the lines $\text{Re}(A) = 0$ and $\text{Im}(A) = 0$. Note how these lines condense along the hole-like localized objects. Part (d) is an enlargement of (c), obtained by interpolation of $\text{Re}(A)$ and $\text{Im}(A)$, focusing on the rightmost hole-like object. In this plot, $x \in [110.6; 111.4]$, $t \in [0; 18]$. The locus of $a_{\min}$ is given by the thin solid line. Note that only three true defects are observed and that the velocity of the object is not zero at these points but somewhat later on. Note also that the trajectory of the objects crosses the $\text{Re}(A) = 0$ and $\text{Im}(A) = 0$ lines at their inflection points.

many partial differential equations of interest found by similar methods can be recast in a simple and powerful setting [16]. This method, involving a ‘projective Riccati sub-system’, shows that solitary wave solutions of equations as diverse as the Korteweg–de Vries and the Kuramoto–Sivashinsky equations can often be written as polynomials of the form:

$$\sum_{i=0}^1 \sum_{j=0}^{P-i} f_{ji}(\xi) (\text{sech}(\kappa\xi))^j (\tanh(\kappa\xi))^i$$

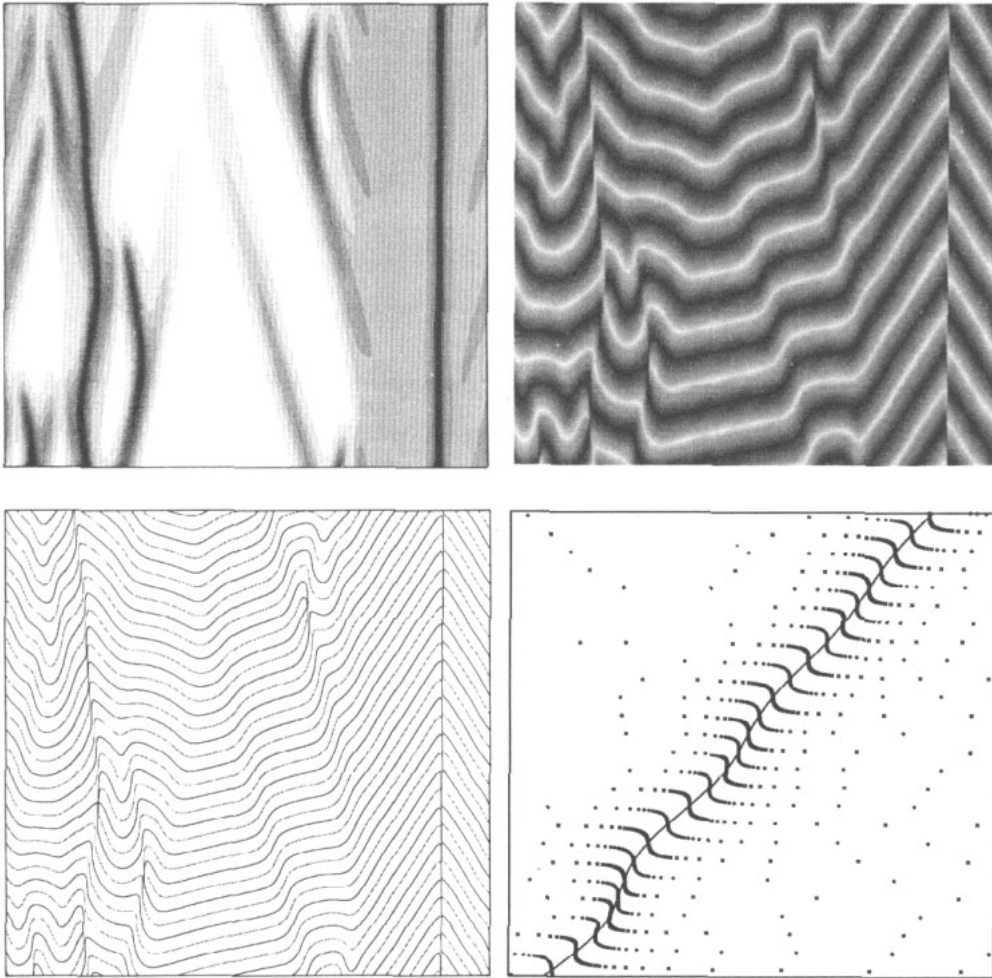


Figure 10. Same as figure 9 but for $b_3 = 0.65$. Time is running upward during $T = 36$. Part (d) focuses on the rightmost hole-like object. In this plot, $x \in [112.2; 113]$, $t \in [0; 36]$. Note that not true defect is observed and that the trajectory of the object passes through the inflection points of the $\text{Re}(A) = 0$ and $\text{Im}(A) = 0$ lines.

where the degree P is given by balancing the highest-order derivative term with the nonlinear terms, and $\xi = x - ct$ with κ, c constant. These results were recently extended to the complex Ginzburg–Landau equation [17], recovering all the Nozaki–Bekki solutions. The known solutions form one family of ‘holes’ [13] plus two special unique solutions [12]. The family of ‘hole’ solutions can be written under the form:

$$|A| = |f_{00} + f_{01} \tanh(\kappa\xi)| \quad \partial_x[\arg(A)] = g_{00} + g_{01} \tanh(\kappa\xi)$$

where the coefficients depend on b_1 , b_3 and c .

These solutions connect two plane waves with wavenumbers k_1 and k_2 by a localized object moving at a velocity $c = (1/b_3 - b_1)(k_1 + k_2)$. Contrary to the shock solution, the hole is characterized by a sharp phase jump σ and a dip of amplitude with $\min(|A|) = a_{\min}$

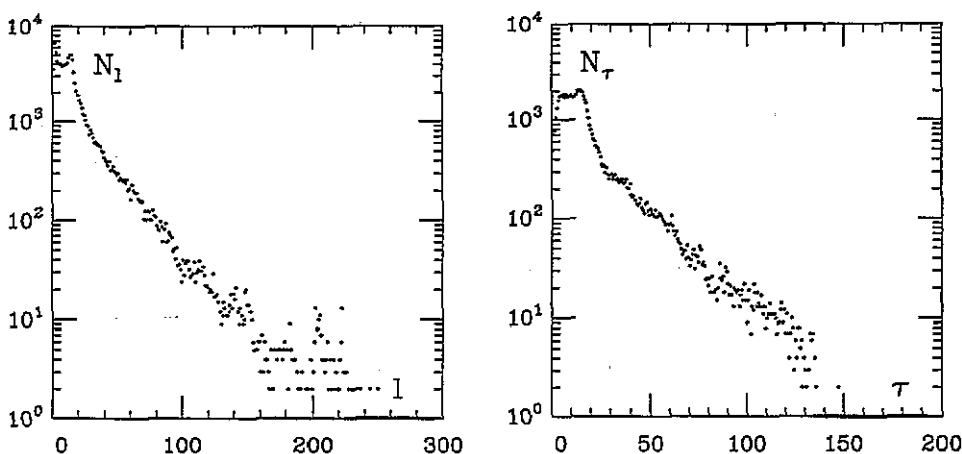


Figure 11. Statistics of the 'laminar' regions in the spatiotemporal intermittency regime observed for $b_1 = 0.5$ and $b_3 = 0.7$ in a system of size $L = 4000$ during $T = 512$ after transients. In this regime (see figure 2), the 'laminar' state consists of linearly stable plane waves with $k \simeq 0$, and $|A| \simeq 1/\sqrt{b_3} = 1.195$. The 'laminar' domains are easily determined by comparing $|A|$ to this value corrected by a tolerance factor. In the present case, a point was declared 'laminar' if $|A| > 1$. This choice was checked to have no influence on the obtained statistics. In this 'far from threshold regime', the run is long enough to produce reliable statistics. Part (a) shows the histogram of the sizes l of the laminar regions cumulated over space and time (bin size $\delta x = 0.976$). Part (b) shows the corresponding histogram for the time lapses τ a point spends in the 'laminar' state (bin size $\delta t = 0.64$). Characteristic scales can be extracted from these exponential distributions. These scales increase, but do not diverge, when approaching the transition line L_2 . The peaks observed for $l \simeq 14$ and $\tau \simeq 15$ correspond to the elementary scales of the objects at the origin of the spatiotemporal disorder (figure 2).

at its centre [18]. This hole constitutes a true defect ($a_{\min} = 0$, $\sigma = \pi$) only for $c = 0$ or, equivalently, for $k_1 + k_2 = 0$. The family can be parametrized by one variable, c , σ , $a_{\min}$ or even $(k_1 + k_2)$, since the asymptotic wavenumbers lie on an ellipse of equation $(k_1 + k_2)^2/K_1^2 + (k_1 - k_2)^2/K_2^2 = 1$, where K_1 and K_2 depend on b_1 and b_3 [13, 19].

The localized objects composing the spatiotemporal intermittency regimes of the complex Ginzburg–Landau equation can be compared to the Nozaki–Bekki solutions in several quantitative ways. Firstly, when the objects are sufficiently isolated from each other (as is the case in figure 3), the plane waves around them reach their 'asymptotic' form and the wavenumbers k_1 and k_2 can be easily measured. The above relationship between c and $(k_1 + k_2)$ has been checked in such cases. The variables k_1 , k_2 and c have been measured for many well-isolated, hole-like objects for several runs at different values of b_1 and b_3 . In all cases, $c = (1/b_3 - b_1)(k_1 + k_2)$ has been verified to an accuracy of a few per cent.

One might object that the relationship between c and $k_1 + k_2$ is not specific enough to characterize the Nozaki–Bekki solutions, so more refined measurements have been performed. Precise evaluations of the *instantaneous* values of $a_{\min}$ and c can be obtained from careful interpolations. In such cases, the hole need not be isolated, since one is not interested in the plane waves around it. Although the Nozaki–Bekki solutions each have a unique, well defined velocity, the comparison can also be extended to objects whose velocity changes substantially in time. One can then investigate whether or not these objects 'follow' the one-parameter family of exact solutions.

The procedure described in figure 13 below reveals that the localized objects usually

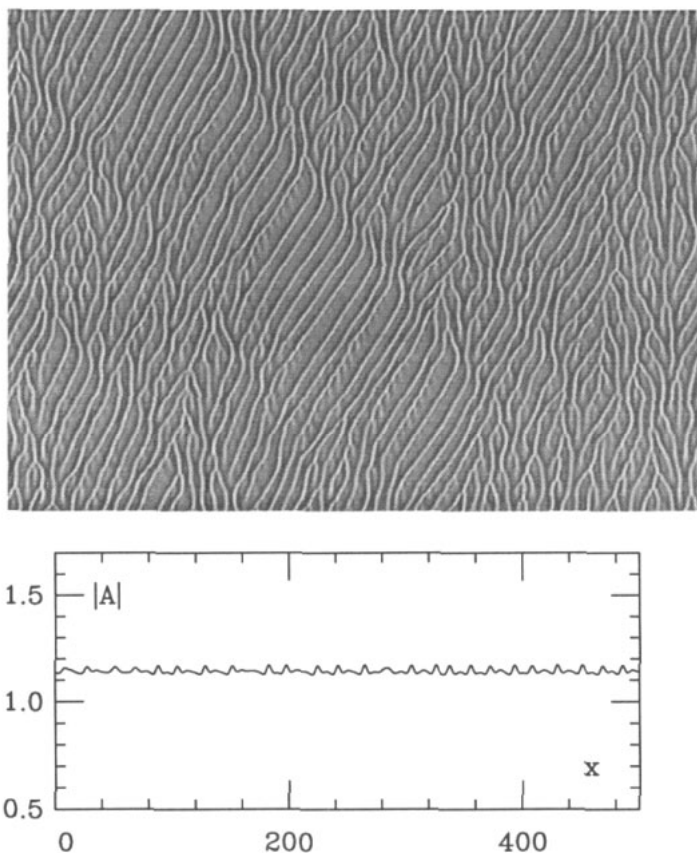


Figure 12. Phase turbulence regime observed for $b_1 = 1$ and $b_3 = 0.75$ in a system of size $L = 1000$ with a winding number of 2π . Part (a) is a spatiotemporal plot of $|A|$ on which time is running upward for $T = 3300$ (note that the timescales are much larger than in the other disordered regimes). Grey scale from $|A| = 1.11$ (black) to $|A| = 1.19$ (white). A representation of $\partial_x[\arg(A)]$ (a more traditional variable for phase turbulence) would produce essentially the same image [26]. Part (b) shows a part of the last solution of the run.

exhibit a simple motion around an exact solution, or even converge onto one. This quantitative analysis confirms that the hole-like objects composing the spatiotemporal intermittency regimes are Nozaki–Bekki solutions, or evolve close to them, in a dynamical way which reveals the structure of their ‘local phase space’.

Finally, it can be readily checked, by mere identification, that the non-propagating, stationary holes present in some of the ‘frozen’ states observed beyond the disordered regimes (figure 6) are Nozaki–Bekki holes.

These results reveal the role played by the Nozaki–Bekki hole solutions in the spatiotemporal intermittency regimes. This first step calls for the search of other exact solutions which would correspond to the wealth of localized objects observed numerically. Richer ansätze than those used in [17] might be able to account for some of them [20]. Nevertheless, it should be noted that such methods, based on Painlevé analysis, do not provide an adequate framework within to look for the ubiquitous ‘branching objects’ observed in the spatiotemporal intermittency regimes (see, e.g. figure 2).

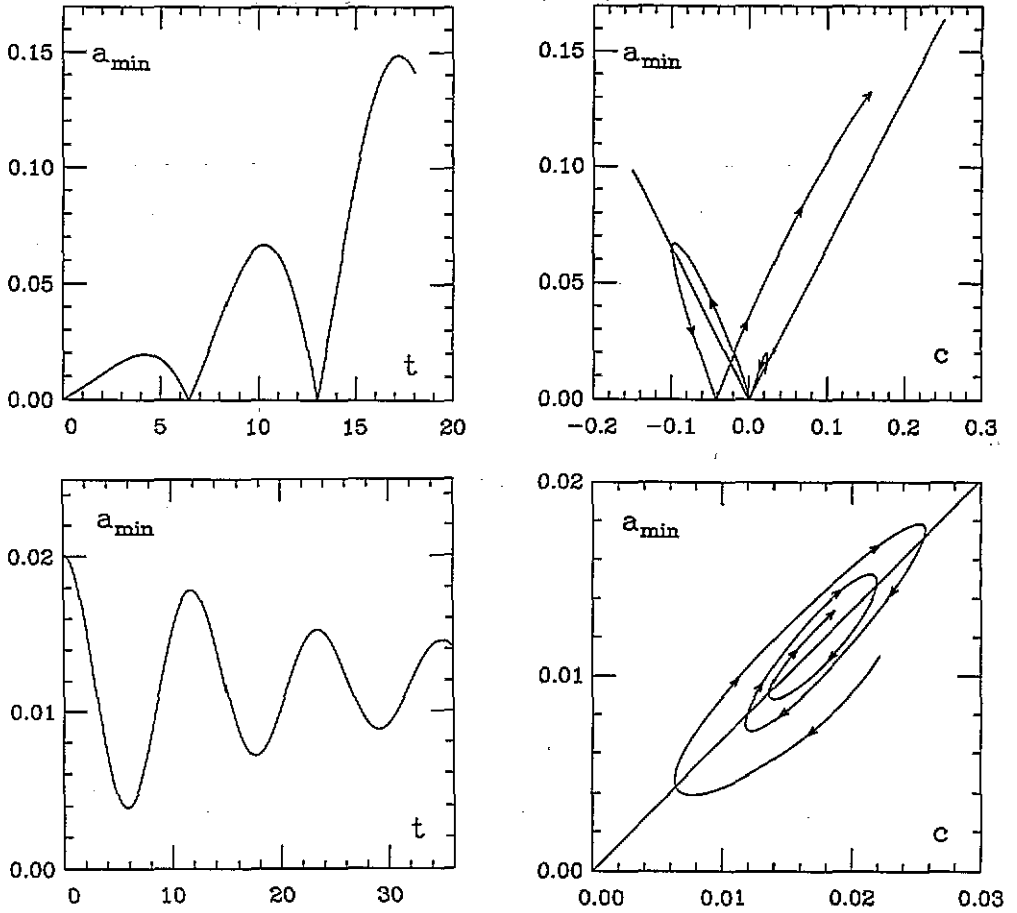


Figure 13. Time series of $a_{\min}$ and variation of $a_{\min}$ with the instantaneous velocity c for the localized hole-like objects studied in figures 9(a), (b) and 10(c), (d). The minimum of $|A|$ was determined at each time step via interpolation of $\text{Re}(A)$ and $\text{Im}(A)$. The instantaneous velocity is given by the derivative of a polynomial interpolation of the trajectories of the objects shown in figures 9(d) and 10(d). The straight lines in (b) and (d) show the representation of the Nozaki-Bekki family of hole solutions in the same variables at the same parameter values.

4. Determining the transition line L_2

Let us now come back to the problem of determining the transition line L_2 in the Benjamin-Feir stable region. In [1], quantities such as correlation functions and density of defects were used to determine the transition line L_1 . Here, the spatiotemporal disorder is far less intense than in those amplitude turbulence regimes [21]. Direct calculations of correlation functions is made extremely difficult by the intermittent character of the dynamical regimes involved, and one has to turn to other ways of determining coherence scales (see [3] and the discussion about figure 11 below). The density of defects have been argued to be of little use, since, firstly, true defects may be extremely difficult to detect (as seen above), and, secondly, such a quantity does not reflect the nature of the spatiotemporal intermittency regimes observed. In this respect, an analysis based on the localized objects described in the previous section seems preferable.

Past studies on spatiotemporal intermittency have been largely devoted to comparing these dynamical regimes with those shown by probabilistic cellular automata such as directed percolation. This was motivated initially by a conjecture of Pomeau suggesting that these two contamination processes might share ‘universal’ features, in particular at the transition point [22]. Statistical characterizations based on a binary reduction of spatiotemporal information were developed, such as the measurement of size and lifetime distributions of the regular, ‘laminar’ regions [3]. This quantitative analysis has been performed for the spatiotemporal intermittency regimes of the complex Ginzburg–Landau equation. Figure 11 shows the histograms of the sizes and lifetimes of the plane wave domains left in between the localized objects carrying the disorder. From these roughly exponential distributions, coherence scales are extracted. When parameter values closer to L_2 are chosen, these coherence scales increase. On the other side of L_2 , these scales are of course infinite. In between, no divergence is observed, as would be the case for a second-order-like transition. Rather, the ‘frozen’ states observed in the transition region prevent this divergence to occur and make L_2 impossible to determine via the coherence scales. In a sense, the deterministic features of the dynamics recover a dominant role here, masking the critical region (for a similar situation in a coupled map lattice model, see [23]). Moreover, for $b_1 \lesssim -0.5$, spatiotemporal intermittency regimes may consist of strongly turbulent patches separated by arrangements of quasi-stable holes (figure 5). The simple statistical characterization also fails in this case, since it accounts for the disordered regions of the system but not for the ‘coherent structures’ which break its spatial extension (see [24] for a discussion of a comparable case).

Given all the difficulties mentioned above, an experimental protocol based on the limit of existence of spatiotemporal disorder was chosen. Line L_2 of figure 1 was determined by scanning the parameter space at fixed values of b_1 and by increasing b_3 by small steps (typically 10^{-2}). At each step, the equation was simulated during a long (fixed) period (typically $T = 7000$), during which the existence of spatiotemporal disorder was checked. When the system reached an asymptotic state with no remaining collision between localized objects or no localized objects at all (homogeneous solution), it was declared on the stable side of L_2 . The whole process was repeated several times for each value of b_1 and shown to produce the same transition points with a satisfactory precision (figure 1). Size-effects were evaluated for some b_1 values by submitting larger systems to the same protocol. They were found to be significant only near the BF line. This is not surprising, since the correlation lengths strongly increase near the BF line and the portion of L_2 above it [1]. These conclusions should not hide, however, that rather large error bars must be kept, due to the essentially qualitative procedure used in determining L_2 . This procedure be seen as a temporary solution, given the failure of the more traditional approaches described above. New, efficient, numerically estimable quantities need to be found to characterize the presence/absence of spatiotemporal disorder in the ‘thermodynamic’ limit of interest here. In this regard, two directions are, I believe, worth investigating: firstly, the signature, in the context of Lyapunov analysis, of the various dynamical regimes involved might be helpful to discriminate between them in an objective and quantifiable manner. The second direction should be toward further identifications of the localized objects composing the spatiotemporal intermittency regimes, toward the study of their stability and, most importantly, of their *interactions*. For example, for the parameter values of figure 5, the Nozaki–Bekki family of hole solutions is stable [31] yet spatiotemporal disorder exists. Standing holes are observed, but their mere stability does not actually provide any help in determining the limit of spatiotemporal disorder. Rather, one would like to be able

to estimate, in such circumstances, the competition between the stable standing Nozaki–Bekki holes and the turbulent phase made of branching objects, i.e. estimate the interactions between these two coexisting phases.

5. Spatiotemporal intermittency in the Benjamin–Feir unstable region

I now turn to the review of the disordered regimes of the complex Ginzburg–Landau equation in the Benjamin–Feir unstable region ($b_1 > b_3$), and discuss to what extent they qualify as spatiotemporal intermittency regimes [25].

The bi-stable region, situated between the BF, L_2 and L_3 lines, satisfies the first requirement for exhibiting spatiotemporal intermittency: two stable states coexist, phase turbulence and an amplitude turbulent state with a finite density of defects [1]. The correspondence with spatiotemporal intermittency situations is completed by noticing that the amplitude turbulence regime is made of localized hole-like objects separating quiescent domains where $|A|$ shows only small, shock-like variations. A closer look at these shock-like objects reveals that they are nothing but the objects composing the phase turbulence regime coexisting at the same parameter values [26]. The phase turbulent state thus appears as the ‘laminar’, absorbing state of spatiotemporal intermittency in which the ‘turbulent’ state (i.e. the hole-like objects) evolves by contamination.

The phase turbulence regimes themselves, stable in the region of the (b_1, b_3) plane situated between the BF, L_1 , and L_3 lines, may present the main features of spatiotemporal intermittency. Preliminary results [26] show that some phase turbulence regimes present the basic features of spatiotemporal intermittency, especially near the L_3 line.

In figure 12, space-time domains of almost-regular trains of propagating shocks are separated by more disordered patches in which the dynamics is strikingly reminiscent of the usual Kuramoto–Sivashinsky model. This rather crude distinction of two ‘local’ states—one regular, one irregular—calls for further analysis in terms of spatiotemporal intermittency.

Elsewhere in the Benjamin–Feir region, i.e. to the left of lines L_1 and L_3 , the amplitude turbulent regimes may resemble those described in section 2. This is particularly true close to the BF line. But, in these regimes, the system lacks a true absorbing state. The phase turbulence dynamics appearing between the hole-like objects (if they are sufficiently far apart), can spontaneously break down [27, 26] and nucleate a defect. Hence these regimes are not strictly of the spatiotemporal intermittency type; the absence of an absorbing state is indeed what accounts for the lack of any critical behaviour in the usual statistical characterization spatiotemporal intermittency performed in [28].

6. Conclusion

The disordered regimes of the complex Ginzburg–Landau equation in the Benjamin–Feir stable region of its parameter plane have been described and argued to qualify as spatiotemporal intermittency regimes. For the first time in a partial differential equation, the ‘laminar’ and the ‘turbulent’ local states at the base of the spatiotemporal intermittency description have been identified. The laminar, absorbing state is formed by linearly stable plane wave solutions. The ‘turbulent’, active state is made of localized objects, the simplest of which are Nozaki–Bekki hole solutions. These objects are the relevant blocks with which a statistical approach of the spatiotemporal intermittency regimes should be built (similar ideas were used by Kishiba *et al* close to the nonlinear Schrödinger equation limit [29]). In

the transition region, ‘freezing’ may occur, stabilizing these objects and producing states with a spatially-disordered structure.

These results complete the general phase diagram first presented in [1]. It should be stressed again that, at this stage, the line L_2 delimiting the spatiotemporal intermittency regimes has been determined crudely. Future work will be devoted to finding adequate quantities to define L_2 .

Although the disordered regimes of the Benjamin–Feir unstable region were reviewed in the present work, some points still deserve more attention. In particular, the phase turbulence regimes and the nature of the transition from phase to amplitude turbulence will be presented elsewhere [26]. Also of interest is the $b_1 \rightarrow \infty$ limit, which should give the asymptotic behaviour of line L_1 , and where the integrable limit given by the nonlinear Schrödinger equation might be useful for analytical treatments.

More time should be devoted to investigating the nature of the transitions between the various phases and in particular to the possibility of any continuous transition leading to a critical point.

At a more general level, the specificity of the one-dimensional case should be assessed by performing a similar systematic exploration of the phase diagram in higher space dimensions where defects are submitted to topological constraints.

Finally, recent works have proposed a number of lines in the (b_1, b_3) plane based on (semi)-analytical studies of stability properties, so that the question of their relationship to the experimental lines L_1 , L_2 , and L_3 naturally arises. For example, the line limiting the existence of ‘convectively stable’ wavenumbers found in [30] lies very close to, if not on, the L_3 line determined experimentally. Another interesting case is the proximity of the L_2 line, for $b_1 \rightarrow -\infty$, to the absolute stability limit of the asymptotic plane waves ‘emitted’ by the symmetric Nozaki–Bekki hole solution [31]. On the other hand, no analytical argument has provided a line close to L_1 . At any rate, the proximity of those lines to each other remains so far mostly coincidental, and a true understanding of why it is so has yet to be given.

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